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Task : Exploring Apache Spark

Lecturer : Dr. Aryati Binti Bakri

Group : 1

Prepared by:

No.	Name	Matric Number
1	Lee Yin Shen	A23CS0236
2	Joanne Ching Yin Xuan	A23CS0227
3	Evelyn Goh Yuan Qi	A23CS0222

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1.0 Business Understanding

In this project, Apache Spark will be used to build an Extraction, Transformation, and Loading (ETL) pipeline. In this project, the original dataset of educational census will be transformed into a structured database for analysis purposes. The dataset from the Brazilian basic education census, which is known as Censo Escolar, will be utilized. This dataset is consisted of school attributes, such as location, administrative type, infrastructure, number of teachers and student enrolment records.

The main problem addressed by this project is the original dataset unable to be analyzed directly. The raw educational census data are stored in a flat form, where textual descriptions and numeric values are incorporated into one file. Consequently, analysis becomes difficult because sorting, grouping, and summarizing actions have to be performed repeatedly before any meaningful information can be extracted. Furthermore, due to huge volume, the CSV file cannot be utilized in business intelligence reporting without any transformations.

Therefore, the main objective of this project is the transformation of the initial school census dataset to a structured form of a structured data warehouse model. This will include ingest the initial CSV file, select relevant attributes, construct dimensional tables and a fact table, and load the processed data into the PostgreSQL database. By using this strategy, an easier way of analysing and querying data would be provided. For instance, analysis of the number of students per state, the number of teachers per location, and presence of school infrastructures in various geographical areas can be made.

The primary stakeholders in the proposed project would include educational analysts, government organizations, school administrators, data engineers, and business intelligence users. The transformed data would be used by educational analysts and policymakers for analysing enrolment pattern and teaching staff. The data would also be used by government organizations in planning for education. In terms of technology, a reusable mechanism for transforming data would be provided by the pipeline for data engineers, while reports and dashboards would be produced by business intelligence users from the structured database.

In summary, the stage of understanding the business requirements indicates that a well-structured ETL process needs to be put in place to increase the accessibility of original education census data. By transforming the original data set into a star schema and storing the

transformed data in PostgreSQL, the accessibility, structure, and analytics of the data have been enhanced.

2.0 Data Understanding

The data set used for this exercise is Censo Escolar 2020 data set, which was sourced from the Brazilian government open data source mentioned in the tutorial. While the original tutorial was intended for multi-year Censo Escolar data, the analysis was done on a single data set of 2020 alone. This choice was made due to the time involved in the processing of multiple-year data sets and the stability issues encountered during execution on the local Windows machine.

The dataset is consisted of information on basic education institutions in Brazil. Four types of attributes are distinguished in the database, namely location attributes, school classification attributes, infrastructure attributes, and numerical educational measures. The location attributes are included with the name, abbreviation, code, name, and code of a state or municipality. Such attributes are considered relevant due to the possibility for education-related data to be analyzed according to geographic areas.

School classification attributes are another category included in the dataset. Examples of classification attributes may be represented by administrative dependency type and school location type. In addition, certain infrastructure attributes were taken into consideration in order to assess whether such facilities in schools as potable water, bathrooms, libraries, computers, Internet, and cafeteria facilities were available. Thus, the described attributes were considered contextual in the analysis.

Numerical attributes associated with teachers and student enrolment are also included in the dataset under consideration. In particular, the attribute QT_DOC_BAS is used to represent the number of teachers in basic education, whereas QT_MAT_BAS is used to represent the number of basic education enrolments. Moreover, attributes such as QT_MAT_INF, QT_MAT_FUND, and QT_MAT_MED are used to show enrolments in various education levels. Since the mentioned numerical attributes could be aggregated, they were used in the fact table.

Several technical characteristics of the dataset were identified during implementation. Firstly, the issue was identified that the dataset employed a semicolon (;) as a delimiter instead of a

comma (,). Secondly, the encoding type should be set to ISO-8859-1 in order for Portuguese characters to be displayed properly. Thirdly, the existence of descriptive value repetitions in state, municipality, school location type and infrastructure indicators was revealed, which indicated the possibility of the dataset being transformed into individual dimension tables.

Thus, prior to transformation, the dataset can be described as a raw relational table. In such a database, descriptive attributes and numerical measures will be present in the same table. While this approach is considered suitable for collecting raw data, it cannot be called a preferred way for presenting information in analytical reports due to the existence of repetitions and numerous attributes in a single structure. Thus, further transformation was performed into a star schema.

Relational Data Model before Transformation

Entity/Table	Attribute Category	Example Attributes	Description
RAW_CENSO_ESCOLAR	Time	NU_ANO_CENSO	Census year
RAW_CENSO_ESCOLAR	Location	NO_UF, SG_UF, CO_UF, NO_MUNICIPIO, CO_MUNICIPIO	State and municipality information
RAW_CENSO_ESCOLAR	School Characteristics	TP_DEPENDENCIA, TP_LOCALIZACAO	Administrative dependency and school location type
RAW_CENSO_ESCOLAR	Infrastructure	IN_AGUA_POTAVEL, IN_BANHEIRO, IN_BIBLIOTECA, IN_COMPUTADOR, IN_INTERNET, IN_REFEITORIO	School infrastructure indicators
RAW_CENSO_ESCOLAR	Measures	QT_DOC_BAS, QT_MAT_BAS, QT_MAT_INF, QT_MAT_FUND, QT_MAT_MED	Teacher and student enrolment counts

In general, this step allowed it to be determined that the Censo Escolar database contains valuable educational data. Nevertheless, raw CSV format is not considered the best choice for performing analytical operations on the data due to repetitions of descriptive attributes and the presence of measurable numerical indicators. Thus, the dataset was considered ready for transformation into a star schema.

3.0 Data Preparation

The data preparation phase included configuring the Apache Spark environment, preparing the dataset, and transforming the raw CSV data into a structured format suitable for analytical processing. The tutorial used the Brazilian Basic Education Census 2020 dataset, which was downloaded and stored locally in CSV format.

Before data processing, some changes had to be made concerning the environment to ensure compatibility with the Windows operating system. As the original tutorial used the Linux system and the Linux commands `wget` and `rm` to process the data, which are not available in the Windows command prompt, the dataset preparation process was modified and executed manually in the local environment.

Apache Spark was then used to read the CSV dataset by using appropriate configuration. The dataset used a semicolon (;) delimiter and required the ISO-8859-1 encoding format to correctly process special characters contained in the data. In addition, schema inference was used to enable automatic detection of appropriate data type for dataset column by Apache Spark during the loading process.

The configuration used to load the dataset into a Spark DataFrame is as follows:

```
# Read CSV data
data_csv = (
    spark
    .read
    .format("csv")
    .option("header", "true")
    .option("inferSchema", "true")
    .option("delimiter", ";")
    .option("encoding", "ISO-8859-1")
    .load("./data/2020.csv")
)
```

Several preprocessing steps were implemented before feeding the data to the dimensional modeling process. Firstly, the necessary columns that should be analyzed were selected from

the raw dataset before transformation to dimension and fact tables. The duplicates within the descriptive records were eliminated by applying the `distinct()` method on Spark to ensure uniqueness of the values stored within the dimension tables. In addition, the casting of data types to numerical columns was done before loading the data into PostgreSQL to make sure compatibility between Apache Spark and the relational database schema.

The following code shows the preprocessing operations performed using Apache Spark, including column selection, data type casting, duplicate removal, primary key generation, and loading the transformed data into PostgreSQL:

```
for table_name, table_config in DIMENSION_TABLES_CONFIG.items():

    print(f"[{datetime.now()}] Writing {table_name}")

    data\
    .select(
        [
            F
            .col(field["field"])
            .cast(field["type"])
            .alias(field["field"])

            for field
            in table_config["fields"]
        ]
    )\
    .distinct()\
    .withColumn(
        "id", F.monotonically_increasing_id()
    )\
    .write\
    .jdbc(
        **POSTGRES_CONFIG,
        table=table_name,
        mode="overwrite"
    )

    print(f"[{datetime.now()}] Wrote {table_name}")
    # Define id as the primary key
    cursor = conn.cursor()
    cursor.execute(
        f"ALTER TABLE {table_name} ADD PRIMARY KEY (id);"
    )
    cursor.close()
    conn.commit()

    print(f"[{datetime.now()}] Added primary key to {table_name}")
    print(f"[{datetime.now()}] Done")
```

The Parquet conversion step was used in the original tutorial. Nevertheless, while implementing this approach, Hadoop NativeIO compatibility problems appeared in the Windows system. The problem was related to writing the Parquet files in Windows. As a result, the dataset was loaded directly from the CSV file without affecting the ETL workflow.

After the preprocessing process, the resulting data was converted into multiple dimension tables and one centralized fact table. Some examples of dimension tables include `dim_local`, `dim_tp_dependencia`, and `dim_tp_localizacao` to contain descriptive information. On the other hand, the `fact_censo_escolar` table was used to contain quantitative measures of education such as the total number of students and teachers. Lastly, the data was successfully loaded into PostgreSQL through JDBC connectivity using Apache Spark.

4.0 Modeling

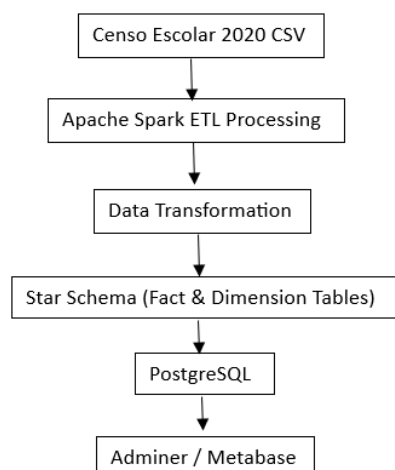
In the modeling stage, efforts were concentrated on converting the preprocessed education dataset into a multi-dimensional data model under the framework of star schema architecture. Star schema was chosen as the basis for the model design due to its popularity in data warehousing and business intelligence applications where it provides an efficient platform for analytics queries and reports. The model consists of fact tables, which hold quantitative data, and dimension tables, which hold qualitative data.

The multi-dimensional data model was built using transformations in Apache Spark according to the steps outlined in the tutorial “Creating a Simple ETL Pipeline With Apache Spark.” Apache Spark was utilized as the primary ETL tool to preprocess the Censo Escolar 2020 dataset, create dimension tables, and build a fact table, which were then loaded into PostgreSQL.

4.1 ETL Architecture

The architecture used for the ETL in the project is based on the distributed computing approach using Apache Spark

The pipeline architecture is shown below:



Apache Spark performed the task of reading the CSV raw data, preprocessing it, transforming its schema, and creating the multidimensional data model. This multidimensional data model was then loaded to the PostgreSQL database. The processed data was analyzed by Adminer and Metabase only for the sake of visualization and inspection.

Only the Censo Escolar 2020 dataset was used in this implementation to reduce processing time while still demonstrating the complete ETL workflow and multidimensional modeling process.

4.2 Multidimensional Data Model

A star schema multidimensional model was created to support analytical querying and business intelligence reporting. The star schema consists of one centralized fact table connected to multiple dimension tables containing descriptive attributes.

4.2.1 Fact Table

The central fact table is: **fact_censo_escolar**

The fact table stores measurable educational statistics used for analysis.

Column Category	Example Columns
Measures	QT_DOC_BAS, QT_DOC_INF, QT_DOC_FUND, QT_DOC_MED, QT_MAT_BAS, QT_MAT_INF, QT_MAT_FUND, QT_MAT_MED
Additional Measures	QT_MAT_BAS_ND, QT_MAT_BAS_BRANCA, QT_MAT_BAS_PRETA, QT_MAT_BAS_PARDA, QT_MAT_BAS_AMARELA, QT_MAT_BAS_INDIGENA
Foreign Keys	ID_DIM_LOCAL, ID_DIM_TP_DEPENDENCIA, ID_DIM_TP_LOCALIZACAO, ID_DIM_IN_BANHEIRO, ID_DIM_IN_BIBLIOTECA, ID_DIM_IN_COMPUTADOR

The measures represent aggregated educational indicators such as total teacher counts (QT_DOC_BAS, QT_DOC_INF, QT_DOC_FUND, QT_DOC_MED) and student enrollment statistics (QT_MAT_BAS, QT_MAT_INF, QT_MAT_FUND, QT_MAT_MED).

Additional measures related to demographic educational statistics, such as QT_MAT_BAS_BRANCA, QT_MAT_BAS_PRETA, QT_MAT_BAS_PARDA, QT_MAT_BAS_AMARELA, and QT_MAT_BAS_INDIGENA, were also included in the fact table as quantitative measures. These features are helpful for demographic analysis of student enrollment data by region and school characteristics. The measures enable users also to analyse the distribution of enrolment by ethnic group and to undertake educational reporting based on infrastructure and location-related information.

4.2.2 Dimension Tables

The following dimension tables were created to provide descriptive context for the fact table.

a) **dim_local**

This dimension stores geographical information related to schools and educational institutions.

Example Attributes
ID_DIM_LOCAL
NO_UF
SG_UF
CO_UF
NO_MUNICIPIO
CO_MUNICIPIO

This dimension supports geographical analysis such as:

- total enrollments by state
- municipality-based reporting
- regional educational analysis

b) **dim_tp_dependencia**

This dimension stores school administrative dependency information.

Example Attribute
ID_DIM_TP_DEPENDENCIA
TP_DEPENDENCIA

This allows analysis based on school administration categories.

c) **dim_tp_localizacao**

This dimension stores school location classification information.

Example Attribute
ID_DIM_TP_LOCALIZACAO

TP_LOCALIZACAO

This dimension supports analytical comparisons between different school location categories.

4.2.3 Infrastructure Dimension Tables

Several additional dimensions related to school infrastructure have also been added as follows:

- dim_in_agua_potavel
- dim_in_banheiro
- dim_in_biblioteca
- dim_in_computador
- dim_in_energia_inexistente
- dim_in_equip_nenhum
- dim_in_esgoto_inexistente
- dim_in_internet
- dim_in_refeitorio

These dimensions store infrastructure-related indicators. Adding infrastructure dimensions allows one to conduct more sophisticated educational analysis, including:

- internet-equipped schools analysis
- facilities comparison in different regions
- infrastructure availability analysis
- resource accessibility assessment

4.3 Star Schema Justification

The decision to use the star schema structure is motivated by its ability to improve query processing. Namely, by isolating descriptive dimensions and measurable facts, queries can be processed in a more efficient way through aggregation of centralized fact data.

The multidimensional schema brings benefits to:

- data queries
- reporting efficiency
- analytical flexibility
- business intelligence applications

The architecture also supports visualization tools such as Metabase by organizing the dataset into a structure suitable for dashboard creation and analytical reporting.

5.0 Evaluation

The ETL pipeline was evaluated based on the successful execution of the extraction, transformation, and loading processes using Apache Spark and PostgreSQL. The evaluation confirmed that the pipeline has been successful in transforming the raw educational dataset into a multidimensional star schema structure that is suitable for analysis.

During the evaluation process, multiple dimension tables and one central fact table were successfully developed in PostgreSQL. The dimension tables that were created include `dim_local`, `dim_tp_dependencia`, `dim_tp_localizacao`, and several infrastructure-related dimensions tables such as `dim_in_biblioteca` and `dim_in_internet`. Also, the `facts_censo_escolar` table successfully stored a large number of educational records, demonstrating Apache Spark's capability to process large-scale educational data efficiently.

Adminer was utilized to verify the accuracy of the ETL results and confirm the correctness of the database structure. The dimension and fact table creation were successfully completed, which proved that the star schema was correctly created and the relationships between fact and dimension tables were correctly implemented in the star schema structure. Figure 5.1 shows the generated tables in PostgreSQL after the ETL process was completed.

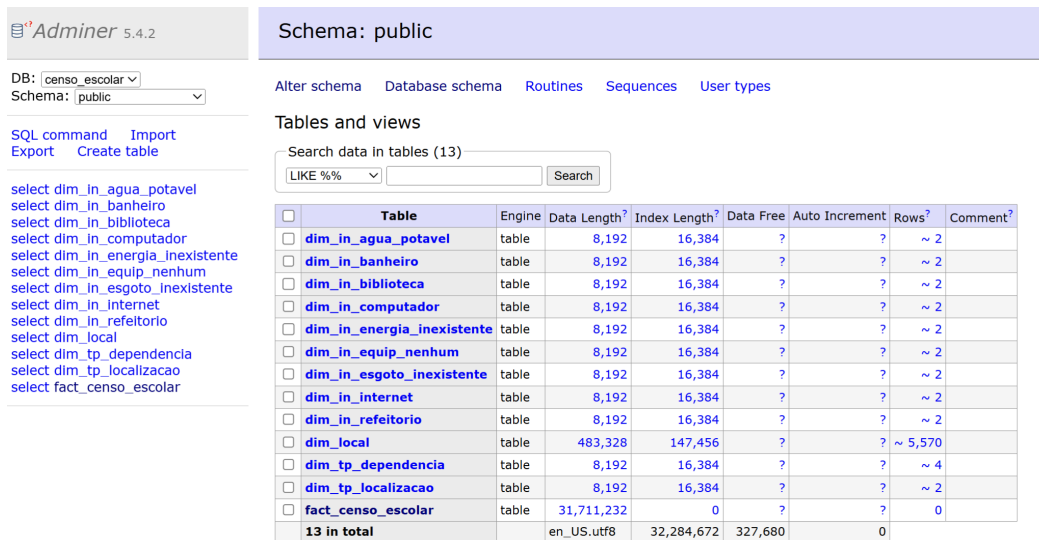


Table	Engine	Data Length?	Index Length?	Data Free	Auto Increment	Rows?	Comment?
☐ dim_in_agua_potavel	table	8,192	16,384	?	?	~ 2	
☐ dim_in_banheiro	table	8,192	16,384	?	?	~ 2	
☐ dim_in_biblioteca	table	8,192	16,384	?	?	~ 2	
☐ dim_in_computador	table	8,192	16,384	?	?	~ 2	
☐ dim_in_energia_inexistente	table	8,192	16,384	?	?	~ 2	
☐ dim_in_esgoto_inexistente	table	8,192	16,384	?	?	~ 2	
☐ dim_in_internet	table	8,192	16,384	?	?	~ 2	
☐ dim_in_refeitorio	table	8,192	16,384	?	?	~ 2	
☐ dim_local	table	483,328	147,456	?	?	~ 5,570	
☐ dim_tp_dependencia	table	8,192	16,384	?	?	~ 4	
☐ dim_tp_localizacao	table	8,192	16,384	?	?	~ 2	
☐ fact_censo_escolar	table	31,711,232	0	?	?	0	
13 in total		en_US.utf8	32,284,672	327,680	0		

Figure 5.1: Fact and dimension tables successfully created in PostgreSQL after ETL execution

During implementation, a compatibility problem occurred at the time of Parquet conversion caused by the limitations of Hadoop NativeIO in the Windows operating system. However, in order to avoid this problem, it was decided to process the dataset directly from CSV files instead of converting it into Parquet format. Despite this limitation, the ETL workflow and star schema implementation were successfully completed without affecting the final output.

Overall, the evaluation showed that the process of the ETL pipeline completion was completed successfully and achieved the objectives. The integration between Apache Spark and PostgreSQL enabled efficient processing, transformation, and storage of large-scale educational data while supporting multidimensional analytical modeling through the star schema architecture.

6.0 Reflection

Lee Yin Shen	From this project, there has been gained some practical knowledge about the pipeline of ETL processes created with Apache Spark. During this process, practical experience of working with Apache Spark in reading the big CSV file, choosing attributes, creating dimensional tables, and creating fact tables has been acquired. Additionally, there has been gained knowledge about PostgreSQL, Adminer, Docker and Metabase, since these tools were used to store, validate and visualize data. Some challenges occurred while implementing the ETL pipeline. Firstly, there were difficulties related with applying the original tutorial in the Windows environment because some commands were intended for Linux. Furthermore, errors were related to downloading the data set, CSV encoding and Parquet conversion. Due to the compatibility issue of Hadoop NativeIO, the Parquet conversion could not be performed. Therefore, it was decided to modify the pipeline by processing the CSV directly. Only the 2020 Censo Escolar data set has been
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	used since the full multi-year data set would take much more time. Overall, practical experience in data pipeline creation has been acquired from this project. Data preprocessing became better understood through this process. This project demonstrated how source data could be transformed into the appropriate format to make analysis possible. Due to using the star model, the data set becomes easier to access with PostgreSQL. For future use, the pipeline could be improved by being executed in Linux or cloud environment, processing all years of data, and creating a more complete dashboard in Metabase.
Joanne Ching Yin Xuan	During the course of implementing the ETL pipeline, various technical difficulties arose due to the complexity of the tasks. The first difficulty involved the large volume of educational data, which slowed down the performance of both Jupyter Notebook and Apache Spark while transforming and loading the dataset. To reduce processing time but still show the whole ETL process, only one database, namely Censo Escolar 2020, was used in the process. Furthermore, the tutorial did not provide specific guidelines on how to set up the development environment, handle errors, and solve issues related to Windows compatibility. Thus, several attempts, troubleshooting, and corrections were required to solve problems related to Apache Spark, Docker, PostgreSQL, and data loading. This indicated the importance of solving problems and being flexible to complete the assignment successfully. Lastly, the project provided an opportunity to learn how to

	implement an ETL pipeline in compliance with the Apache Spark process illustrated in the tutorial. Skills were learned in data extraction, transformation, multidimensional modeling, and data integration into a PostgreSQL database using the ETL process. In addition, more skills were gained in Docker configuration, database validation using Adminer, and analysis through Metabase.
Evelyn Goh Yuan Qi	This tutorial provided practical experience in implementing ETL pipelines using Apache Spark and PostgreSQL in a local Docker setup. The skills related to data preparation on a massive scale using Apache Spark DataFrames were reinforced via techniques such as schema inference, selecting columns, data type casting, eliminating duplicates, and transforming data multidimensionally into dimension and fact tables for creating a star schema model. Furthermore, knowledge about the integration between Apache Spark and PostgreSQL using JDBC connections was improved when transferring data from the ETL pipeline into a relational database management system. Some technical problems also occurred during the process, especially regarding the incompatibility between Hadoop NativeIO and the Windows operating system during the data transformation from CSV to Parquet format. These issues were addressed by adjusting the ETL pipeline design to process the data from the source CSV file without converting it to Parquet format. Ultimately, better understanding of the technical aspect in ETL pipeline development, multidimensional data modeling, and massive-scale data processing using Apache Spark technologies was gained through this tutorial.

